

CENTENNIAL, CO, USA

WMO: 724666

Lat: 39.570N

Lon: 104.849W

Elev: 1793

StdP: 81.56

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93067

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.3	-13.9	-22.1	0.6	-9.3	-19.7	0.8	-4.9	12.1	5.4	10.7	4.0	2.3	40	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	33.4	15.2	32.2	15.0	30.8	14.8	18.1	26.7	17.3	25.7	16.6	25.3	4.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	14.2	20.1	14.8	13.1	19.8	13.7	12.2	19.6	58.8	26.4	55.6	25.1	53.1	25.1	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	8.1	-21.7	36.3	2.4	1.1	-23.4	37.1	-24.8	37.7	-26.1	38.4	-27.9	39.2		
(5)			-22.0	19.3	2.4	1.2	-23.7	20.2	-25.1	20.9	-26.4	21.6	-28.2	22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	0.1	0.7	5.0	8.2	13.3	19.5	23.0	21.7	17.4	10.2	4.6	-0.1
	DBStd	9.62	5.86	6.17	5.75	5.27	4.81	3.86	2.70	2.66	4.51	5.38	5.88	6.10
	HDD10.0	1418	308	263	171	93	24	1	0	0	6	62	174	315
	HDD18.3	3286	565	494	414	305	165	32	3	6	67	253	411	572
	CDD10.0	1549	1	2	16	39	127	285	402	364	229	69	13	1
	CDD18.3	374	0	0	0	1	9	66	147	112	39	2	0	0
	CDH23.3	4477	0	0	2	14	179	915	1612	1149	541	64	1	0
	CDH26.7	1763	0	0	0	1	38	366	734	450	168	6	0	0
Wind	WSAvg	3.7	3.5	3.8	4.0	4.3	3.9	3.9	3.7	3.5	3.4	3.5	3.5	3.4
Precipitation	PrecAvg	428	14	18	32	55	58	48	55	54	36	28	21	15
	PrecMax	613	32	56	108	140	149	111	111	115	174	71	68	71
	PrecMin	248	2	0	0	3	15	12	8	8	0	0	6	0
	PrecStd	86	8	13	26	30	33	27	30	27	33	19	15	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	19.2	23.0	25.1	30.1	34.5	36.0	34.0	32.7	27.5	22.5	17.7
		MCWB	5.7	6.4	8.8	9.6	12.9	14.7	15.9	15.1	14.6	11.4	7.9	5.8
	2%	DB	14.4	16.2	20.3	22.8	27.5	32.4	33.9	32.5	30.2	25.2	19.6	14.5
		MCWB	4.2	5.0	7.2	8.9	12.4	14.4	15.5	15.0	13.5	10.3	6.9	4.3
	5%	DB	11.9	13.4	17.9	21.1	25.3	30.9	32.5	31.0	28.6	22.9	17.3	12.0
		MCWB	3.1	3.8	6.3	8.2	11.6	14.1	15.4	15.0	13.0	9.6	6.1	3.2
	10%	DB	9.1	10.7	15.4	18.4	22.9	28.8	31.1	29.3	27.0	20.2	14.4	9.1
		MCWB	1.9	2.6	5.2	7.4	11.2	13.9	15.3	15.0	12.8	8.8	4.9	2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	7.0	10.4	11.3	15.4	18.3	19.6	19.4	17.3	13.5	8.6	6.3
		MCDB	15.9	17.8	19.5	21.1	25.8	28.0	28.6	27.6	26.3	22.4	20.0	16.7
	2%	WB	4.7	5.4	7.9	10.0	14.0	16.8	18.3	18.0	15.7	11.7	7.6	4.7
		MCDB	13.5	15.3	18.7	19.8	23.1	26.1	27.0	26.1	24.8	21.4	19.0	13.8
	5%	WB	3.3	4.1	6.7	8.9	12.9	15.7	17.4	17.2	14.7	10.5	6.4	3.4
		MCDB	11.1	12.7	16.9	18.8	21.9	25.3	25.8	25.3	23.9	20.3	16.6	11.3
	10%	WB	2.1	2.8	5.5	7.9	11.8	15.0	16.7	16.4	13.8	9.4	5.2	2.0
		MCDB	8.9	10.2	14.5	17.4	20.7	24.8	25.4	24.7	23.1	18.7	13.8	8.9
Mean Daily Temperature Range	5% DB	MDBR	13.0	13.1	13.9	13.4	13.6	15.0	14.7	14.2	14.7	14.2	13.7	13.0
		MCDBR	16.0	16.8	17.2	17.3	16.9	17.8	17.0	16.3	17.2	18.0	17.4	16.2
		MCWBR	8.6	8.7	8.4	7.6	6.6	6.0	5.2	4.9	5.7	7.5	8.5	8.8
	5% WB	MCDBR	15.3	16.0	16.6	15.8	15.1	15.2	14.1	13.9	14.8	15.7	16.6	15.6
		MCWBR	8.5	8.7	8.4	7.6	6.7	6.2	5.3	5.2	5.5	7.0	8.5	8.8
Clear-Sky Solar Irradiance	taub	0.236	0.242	0.263	0.291	0.310	0.319	0.347	0.337	0.301	0.267	0.247	0.238	
	taud	2.632	2.626	2.572	2.507	2.515	2.503	2.466	2.480	2.560	2.643	2.622	2.624	
	Ebn at Noon	972	1006	1006	987	967	954	923	927	949	961	947	944	
	Edh at Noon	65	76	90	103	104	106	109	105	90	74	65	61	
All-Sky Solar Radiation	RadAvg	2.61	3.61	4.81	5.70	6.29	7.14	6.76	6.07	5.27	3.91	2.81	2.29	
	RadStd	0.11	0.19	0.36	0.39	0.49	0.33	0.30	0.21	0.27	0.23	0.16	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	+34
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62	N/A

Nomenclature: See separate page